

Saddle-node bifurcation of first-passage-time densities induced by a directed shortcut

Xiaoxiao Zhouyi^{1,*} and Luca Giuggioli¹

¹*School of Engineering Mathematics and Technology,
University of Bristol, Bristol BS8 1TW, United Kingdom*

(Dated: July 1, 2026)

First-passage-time (FPT) densities in shortcut-assisted search can be bimodal: a fast “direct” peak and a slow “indirect” peak. We show that this bimodality is the observable-level signature of a *saddle-node bifurcation* that can be solved exactly. Modeling a directed shortcut to an absorbing target as a rank-one killing defect on a lazy lattice—whose diffusive limit is Brownian motion on an interval with an interior δ -sink of strength b at fractional position θ —we show that the second FPT peak is born and annihilated at a fold of a *signed* spectral mixture, governed by the compact criterion $S_1 = S_2 = 0$, where $S_n(\tau) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}$ and $f' = -S_1$, $f'' = S_2$. The double peak exists if and only if $0 < b < b_c(\theta)$ (for the source-started convention $\xi = \theta$ used throughout), with a computable boundary $b_c(\theta)$ that is symmetric about $\theta = \frac{1}{2}$, has a minimum near $\theta \simeq 0.381$, and diverges as $b_c \sim 0.789 / \min(\theta, 1 - \theta)$ near the endpoints; at least three sign-alternating modes are required, and the fold obeys the generic saddle-node normal-form scaling $\tau_+ - \tau_- \propto (b_c - b)^{1/2}$, $\Delta f \propto (b_c - b)^{3/2}$. The local fold mechanism is not lattice-specific: it follows model-independently from a low-rank killing defect (Woodbury identity), it generalizes to multiple shortcuts (two shortcuts create a *third* peak organized by a cusp catastrophe, which we locate), and it persists on finite two-dimensional lattices. The threshold b_c is a time-domain morphology fold, *not* a spectral exceptional point. The result recasts a known phenomenology as a predictive, exactly solvable bifurcation with a measurable threshold and universal scaling, testable in feedback-controlled colloidal search and in engineered transport networks.

I. INTRODUCTION

The first-passage time (FPT) to a target is a central observable of stochastic transport, search and reaction kinetics [1–3]. Its full distribution $f(t)$, rather than the mean first-passage time (MFPT) alone, controls the timing statistics of diffusion-limited reactions, of intracellular delivery, and of search under deadlines, and it is increasingly accessible experimentally through single-particle tracking and feedback-controlled colloidal platforms [9, 10].

A recurring and striking feature of $f(t)$ is *multimodality*. When a searcher has access to a “shortcut”—a relocation channel, a long-range rewired edge, or an active-transport route—the FPT density can become bimodal: fast trajectories that use the shortcut form an early peak, while slow trajectories that explore the domain form a later peak, provided the two time scales are well separated [6]. Random walks on small-world and shortcut-rich networks likewise exhibit shortcut-induced temporal structure in their first-passage statistics [8]. Multimodal capture-time densities are further known for interior partially absorbing traps when several traps are strategically arranged [9, 10], and exact spatiotemporal solutions of confined lattice random walks with defects are by now a mature program [4, 5].

These works *exhibit* multimodality; they do not, however, classify or predict *when* a second peak exists as a control parameter is varied, nor identify the transition as

a bifurcation with a sharp, computable threshold. The distinction matters: the presence of a second arrival-time scale is a qualitative, controllable property of the distribution that is invisible to the MFPT and to the spectral gap, and its appearance and disappearance is, we argue, a genuine catastrophe of the observable.

Here we supply exactly that classification for a minimal directed shortcut, modeled as a rank-one killing (non-Hermitian) defect on an absorbing lattice. Our contributions are:

1. An *exact solution* (Sec. II): the FPT density is a signed spectral mixture whose amplitudes we derive in closed form from the defect resolvent, both on the finite lattice (Montroll determinant, App. A) and in the diffusive limit (App. B).
2. A *saddle-node fold criterion* (Sec. III): the birth and death of the second FPT peak is a fold governed by the compact local criterion $S_1 = S_2 = 0$ in signed spectral moments, with $\Phi' = -S_1$, $\Phi'' = S_2$. For a start at the source ($\xi = \theta$), the second peak exists precisely when $0 < b < b_c(\theta; \xi)$, a boundary we determine and certify numerically.
3. A *computable phase boundary* $b_c(\theta)$: symmetric about $\theta = \frac{1}{2}$, minimized near $\theta \simeq 0.381$, with the endpoint law $b_c \sim 0.789 / \min(\theta, 1 - \theta)$; a *minimal-mode theorem* (at least three sign-alternating modes are required); and the universal normal-form scaling $\tau_+ - \tau_- \propto (b_c - b)^{1/2}$, $\Delta f \propto (b_c - b)^{3/2}$.
4. A *universality* argument (Sec. IV): for reversible (symmetrizable) low-rank killing the fold is model-

* xiaoxiao.zhouyi@bristol.ac.uk

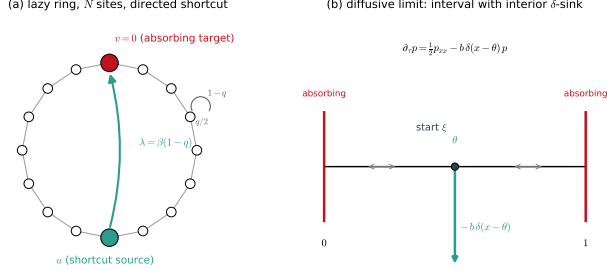


FIG. 1. Model. (a) Lazy random walk on a ring of N sites (stay $1 - q$, hop $q/2$) with an absorbing target $v = 0$ and a directed shortcut $u \rightarrow v$ of rate $\lambda = \beta(1 - q)$; deleting v leaves a rank-one killing defect at u . (b) Diffusive limit: Brownian motion on $[0, 1]$ with absorbing ends and an interior δ -sink of strength b at θ , started at ξ .

independent (Woodbury identity), generalizes to multiple shortcuts (two shortcuts create a *third* peak organized by a cusp catastrophe, which we locate), and persists on finite two-dimensional lattices.

5. A *physical embedding and non-Hermitian clarification* (Sec. V): the shortcut is a localized target-delivery channel; the threshold b_c is a time-domain morphology fold, *not* a spectral exceptional point.

We are careful to separate what is new from what is not. Shortcut-induced bimodality and interior-trap multimodality are known [6, 9, 10]; the lattice Green-function/defect method is Giuggioli's [4, 5]; and Ref. [7] concerns bimodality of a splitting probability $P(\omega)$, not of $f(t)$. Our contribution is the exactly solvable *bifurcation-theoretic* account: the fold criterion, the predictive threshold $b_c(\theta)$, and the universality of the mechanism.

II. MODEL AND EXACT SOLUTION

A. Lazy ring with a directed shortcut

Consider a lazy random walk on a ring of N sites [Fig. 1(a)]: at each step the walker stays with probability $1 - q$ and hops to each neighbor with probability $q/2$. Site $v = 0$ is an absorbing target. A directed shortcut at a source site u diverts probability $\lambda = \beta(1 - q)$ from the u self-loop onto the directed edge $u \rightarrow v$; upon taking it the walker is absorbed. Removing the absorbed state leaves a transient (sub-stochastic) generator on sites $1, \dots, N - 1$ that is a Dirichlet path with a rank-one diagonal killing defect at u ,

$$Q_b = Q_0 - b |u\rangle\langle u|, \quad (1)$$

so probability leaks irreversibly from u into the target. Although the shortcut is directed, the transient

block remains symmetric—the directedness enters only through the (one-way) absorption vector, not the survival dynamics—so its spectrum is real. This is the crucial structural fact that makes the problem exactly solvable while retaining genuinely non-Hermitian (open, probability-losing) character.

B. Exact generating function and Montroll determinant

The FPT generating function is rational with an exact Montroll-type determinant. Applying the matrix-determinant lemma to the rank-one defect (1) and evaluating the resulting path-resolvent element in closed form (App. A) gives, with U_m the Chebyshev polynomials of the second kind, $y = 1 + (z^{-1} - 1)/q$ the spectral variable, and $a = q/\lambda$ the dimensionless inverse shortcut strength, the characteristic determinant for a source at u ,

$$D_u(y) = a U_{N-1}(y) + 2 U_{u-1}(y) U_{N-u-1}(y). \quad (2)$$

The FPT probability is then a finite sum of geometric modes,

$$F(t) = \sum_j B_j s_j^{t-1}, \quad s_j = 1 - q + q y_j, \quad (3)$$

where y_j are the roots $D_u(y_j) = 0$ and the residues are $B_j = q \mathcal{N}_{r,u}(y_j)/D'_u(y_j)$ with r the start site and $\mathcal{N}_{r,u}$ the piecewise numerator of App. A. The antipodal case $u = N/2 \equiv L$ reduces (2) to $D \propto a T_L + U_{L-1}$ (with T_L the Chebyshev polynomial of the first kind), recovering the reflection-symmetric problem as a special case of the general rank-one theory.

C. Diffusive limit

Writing $\theta = u/N$ for the fractional shortcut position, $b = \beta(1 - q)N/q$ for the rescaled sink strength, $\tau = qt/N^2$ for diffusive time, and expanding $s = 1 - q + q \cos(2w/N)$, the walk converges to Brownian motion on $[0, 1]$ with absorbing ends and an interior δ -sink [Fig. 1(b)],

$$\partial_t p = \frac{1}{2} p_{xx} - b \delta(x - \theta) p, \quad p(0) = p(1) = 0. \quad (4)$$

Its spectrum is fixed by the transcendental determinant

$$D_\theta(k; b) = k \sin k + 2b \sin(k\theta) \sin(k(1 - \theta)) = 0, \quad (5)$$

with $k = 2w$ and eigenvalues $\mu_j = k_j^2/2 = 2w_j^2$; for $\theta = \frac{1}{2}$ this collapses to $\tan w = -2w/b$. This interval δ -sink (point-interaction) eigenvalue condition is the standard point-sink spectrum [9, 10]; our exact-solution novelty is not this spectrum but the finite- N directed-shortcut Montroll determinant D_u [Eq. (2)] and the bifurcation classification developed below. The FPT density from a start at $\xi = r/N$ is the signed spectral mixture

$$\Phi_{\xi,\theta}(\tau; b) = \sum_j G_j e^{-\mu_j \tau}, \quad (6)$$

whose amplitudes we derive from the defect resolvent (App. B) as

$$G_{\xi,\theta} = 2w^2 \frac{\phi_{k,\theta}(\xi) I(k, \theta)}{J(k, \theta)}, \quad k = 2w, \quad (7)$$

where

$$\phi_{k,\theta}(x) = \begin{cases} \sin(k(1-\theta)) \sin(kx), & x \leq \theta, \\ \sin(k\theta) \sin(k(1-x)), & x \geq \theta, \end{cases} \quad (8)$$

and $I = \int_0^1 \phi dx$, $J = \int_0^1 \phi^2 dx$ have the closed forms of App. B. Modes with $\sin(n\pi\theta) = 0$ do not couple to the sink and contribute the free Dirichlet term $G_n^{\text{node}} = n\pi[1 - (-1)^n] \sin(n\pi\xi)$ (even n vanish). We have verified (7), the crucial normalization identity $J = \sin(k) D_k/(4b)$, and the full master curve (affected plus node modes) against exact finite- N residues to a relative error $\sim 10^{-5}$ with clean $O(N^{-2})$ convergence (App. B, App. E).

III. SADDLE-NODE FOLD CRITERION AND PHASE BOUNDARY

A. Signed spectral moments and the fold criterion

Define the signed spectral moments

$$S_n(\tau; b) = \sum_j G_j \mu_j^n e^{-\mu_j \tau}, \quad (9)$$

so that $\Phi' = -S_1$ and $\Phi'' = S_2$ [differentiate (6)]. An interior extremum of the FPT density is a zero of S_1 ; the second peak is a valley-peak pair, and it is born or annihilated when these two extrema merge—a *saddle-node* fold of the density's critical points,

$$\boxed{S_1(\tau_c, b_c) = 0, \quad S_2(\tau_c, b_c) = 0}, \quad (10)$$

with nondegeneracy $S_3(\tau_c, b_c) \neq 0$ and $\partial_b S_1(\tau_c, b_c) \neq 0$. No positivity is assumed—the G_j are genuinely signed—so (10) is the correct fold criterion for a signed exponential mixture, and it is precisely the coincidence of a root and a critical point of the generalized (signed) exponential sum S_1 . Throughout we adopt the source-started convention $\xi = \theta$; more generally the amplitudes G_j , and hence the boundary, depend on the start, so that $b_c = b_c(\theta; \xi)$. The local fold (10) is exact; the global statement—that for $\xi = \theta$ the second peak exists precisely for $0 < b < b_c(\theta)$, the double-peak set being a single connected interval in b with no re-entrant window over the scanned range—we certify numerically (App. E), rather than prove analytically.

B. The existence boundary $b_c(\theta)$

Eliminating τ_c between the two equations in (10) yields a scalar boundary $b_c(\theta)$ [Fig. 2(a)]. It is symmetric,

$b_c(\theta) = b_c(1-\theta)$, has a minimum at $\theta \simeq 0.381$ ($b_c \simeq 2.16$), and obeys the endpoint law

$$b_c(\theta) \sim \frac{0.7890261736}{\min(\theta, 1-\theta)}, \quad \tau_c \sim 0.1579221011 d^2, \quad (11)$$

with $d = \min(\theta, 1-\theta)$, obtained from the half-line boundary-layer problem of App. C. In the antipodal case

$$b_c(\frac{1}{2}) = 3.0764323604, \quad \tau_c = 0.03836305186, \quad (12)$$

with the near-antipodal expansion $b_c(\frac{1}{2} + \epsilon) = 3.0764323604 - 133.107 \epsilon^2 + O(\epsilon^4)$ (App. C). The two limits are physically transparent: as $b \rightarrow 0$ the spectrum reverts to the plain Dirichlet spectrum (no sink, no second peak), while as $b \rightarrow \infty$ the δ -sink becomes an absorbing cut and, for a start at the source $\xi = \theta$, the density collapses to $\Phi \rightarrow \delta(\tau)$ (again no interior second peak)—so a second peak can exist only in the finite window $0 < b < b_c$.

C. Normal-form scaling

Near the fold the two merging extrema split with the generic saddle-node exponents

$$\tau_+ - \tau_- \propto (b_c - b)^{1/2}, \quad \Phi(\tau_+) - \Phi(\tau_-) \propto (b_c - b)^{3/2}, \quad (13)$$

which we confirm numerically [Fig. 2(c)]; in the antipodal case the prefactors are $\tau_+ - \tau_- \simeq 0.0247518 (b_c - b)^{1/2}$ and prominence $\simeq 0.357444 (b_c - b)^{3/2}$. The observation of these exponents certifies the transition as a fold rather than a smooth spectral crossover, and they constitute a system-independent experimental prediction (Sec. V).

D. Minimal mode count

A two-mode fold is impossible, even with signs. Writing $A_j = G_j \mu_j e^{-\mu_j \tau}$, $S_1 = 0$ forces $A_2 = -A_1$, whence $S_2 = (\mu_1 - \mu_2) A_1 = 0$ gives $A_1 = A_2 = 0$, a contradiction. Thus *at least three* modes are required, and the fold ratio $A_1 : A_2 : A_3 = (\mu_3 - \mu_2) : (\mu_1 - \mu_3) : (\mu_2 - \mu_1)$ forces *alternating signs*, $G_1 > 0, G_2 < 0, G_3 > 0, \dots$, exactly the sign pattern the true affected amplitudes possess (App. B). Retaining K affected modes in the antipodal case gives the truncated thresholds $b_c^{(2)}$ (none), $b_c^{(3)} = 3.046$, $b_c^{(5)} = 3.0764$, converging rapidly to (12).

IV. UNIVERSALITY

A. Model-independence

The fold (10) is not special to the ring. For any absorbing Markov generator Q_0 , a rank- m killing defect $Q_b = Q_0 - UBU^\top$ gives, by the Woodbury identity,

$$\det(sI - Q_b) = \det(sI - Q_0) \det[I_m + BW(s)], \quad (14)$$

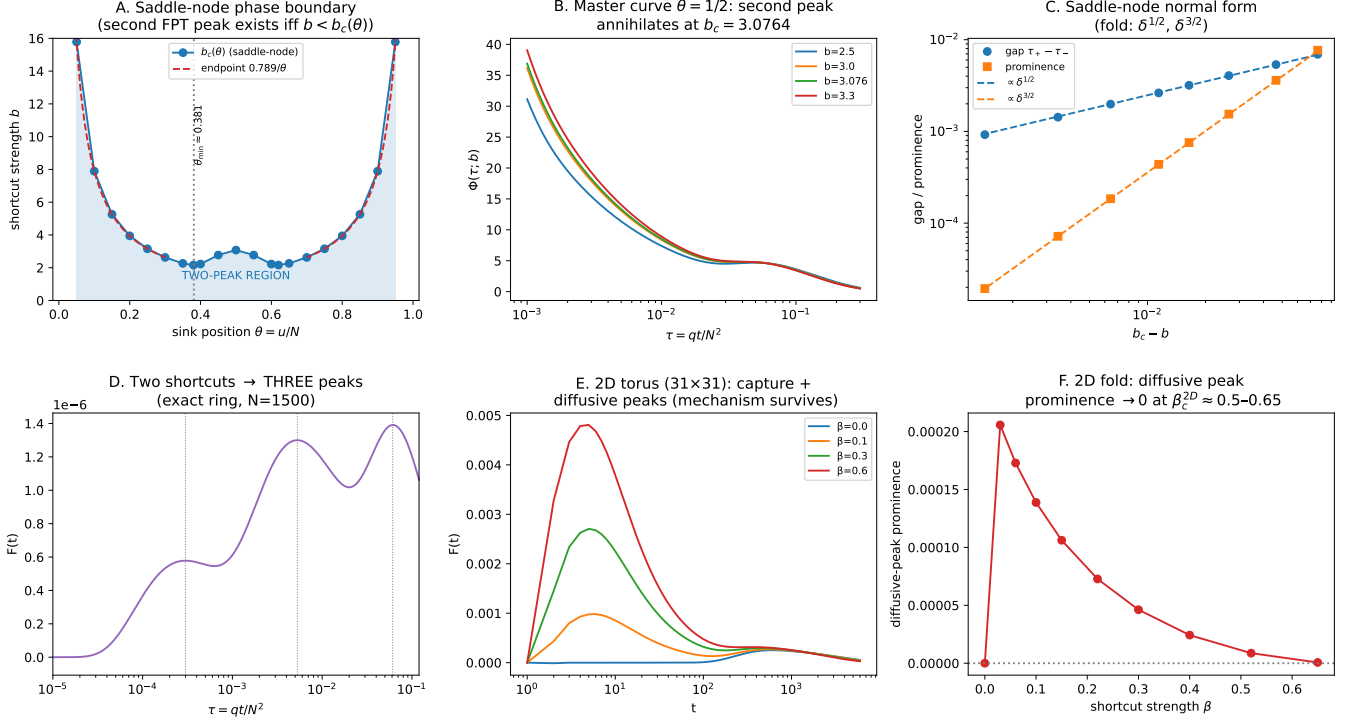


FIG. 2. Results. (a) Saddle-node phase boundary $b_c(\theta)$: the second FPT peak exists for $0 < b < b_c(\theta)$; symmetric, minimum at $\theta \simeq 0.381$, with the endpoint law $0.789/\min(\theta, 1-\theta)$ (dashed, both branches). (b) Master curve $\Phi(\tau; b)$ at $\theta = \frac{1}{2}$ as b crosses $b_c = 3.0764$. (c) Normal-form scaling: gap $\propto (b_c - b)^{1/2}$ and prominence $\propto (b_c - b)^{3/2}$. (d) Two shortcuts produce a third peak (exact ring, $N = 1500$). (e) 2D torus (31×31): capture and diffusive peaks. (f) 2D fold: the diffusive-peak prominence vanishes at $\beta_c^{2D} \simeq 0.55-0.65$.

with $W_{ab}(s) = \langle u_a | (sI - Q_0)^{-1} | u_b \rangle$. For *reversible* (symmetrizable) transient dynamics with diagonal low-rank killing—as for the symmetric ring block here— Q_b has a real spectrum and the FPT density is again a real signed spectral mixture, whose second-peak birth/death is the fold $S_1 = S_2 = 0$. (For non-reversible directed transient dynamics the determinant identity still holds, but the real modal expansion, and hence the fold classification, requires additional hypotheses.) The double-peak region is an *open* set (structurally stable by the implicit-function theorem), hence not fine-tuned. The honest universal statement is not “every shortcut creates a peak”—false in general—but “whenever a shortcut-induced peak is born or annihilated generically, the mechanism is a fold of the signed spectral mixture, independent of the underlying geometry.”

B. Multiple shortcuts and the cusp

For two shortcuts at $\theta_1 = \alpha < \theta_2 = \eta$ (positions; β is reserved for the shortcut rate) with strengths b_1, b_2 , the

secular determinant is (App. D)

$$D_{12}(k) = k \sin k + 2b_1 \sin(k\alpha) \sin(k(1-\alpha)) + 2b_2 \sin(k\eta) \sin(k(1-\eta)) + \frac{4b_1 b_2}{k} \sin(k\alpha) \sin(k(\eta-\alpha)) \sin(k(1-\eta)), \quad (15)$$

the last term being a genuine two-defect interaction that vanishes as the sources coalesce (giving the additive rule $b = b_1 + b_2$). Two shortcuts can produce a *third* FPT peak: at $(\alpha, \eta, b_1, b_2) = (0.38, 0.48, 1.35, 0.14)$, $\xi = 0.463$, the exact ring ($N = 1500$) shows three peaks at $\tau \simeq 3.0 \times 10^{-4}, 5.3 \times 10^{-3}, 6.1 \times 10^{-2}$ [Fig. 2(d)], corresponding to capture via the near shortcut, capture via the far shortcut, and diffusive arrival. The triple-peak wedge in (b_1, b_2) is organized by a *cusp* $S_1 = S_2 = S_3 = 0$; we locate it at

$$(\alpha, \eta, b_1, b_2, \tau_c) = (0.38, 0.489580512568, 1.35, 1.08389, 0.0012366), \quad (16)$$

with $S_4 \neq 0$ and unfolding Jacobian $\det \partial(S_1, S_2)/\partial(b_1, b_2) = -6.9 \times 10^7 \neq 0$, a genuine codimension-two cusp (App. D). We note that the cusp

sits slightly off the triple-peak example ($\eta \approx 0.48958$, not 0.48, which carries only ordinary folds), so the two statements are consistent but distinct. Thus m shortcuts turn the single fold into fold hypersurfaces in b -space, with open regions of two and three peaks, organized by a hierarchy of catastrophes [11]—all exactly solvable.

C. Two dimensions

On an $L \times L$ torus with an absorbing target and a directed shortcut, the same algebra holds [Eq. (14) with the 2D lattice resolvent]. Numerically (at $L = 31$), a single diffusive peak at $\beta = 0$ acquires a shortcut-induced capture peak; the two coexist over a window and the diffusive peak's prominence falls to zero near $\beta_c^{2D} \simeq 0.5$ – 0.65 [Fig. 2(e,f)], the same fold morphology as in one dimension. One caveat is essential: a single-site sink is marginal in 2D (its self-Green function is logarithmically divergent), so the *bare* threshold is expected to drift as $b_c(L) \sim 1/\log L$; the large- L continuum point-sink limit therefore requires logarithmic renormalization, and we leave a size-independent 2D scaling test (the analogue of (13)) to future work. We thus present the 2D result as finite-lattice evidence that the mechanism is not one-dimensional, not as a proof of 2D universality.

D. Robustness

The fold persists under any perturbation that preserves the nondegeneracy $S_3 \neq 0$, $\partial_b S_1 \neq 0$: to first order $\delta b_c = -(\partial_\varepsilon S_1 / \partial_b S_1) \delta \varepsilon$ for a perturbation ε . Concretely it survives weak hop-rate disorder (which can even help, by lifting accidental dark modes), small shifts of the source or start, and the laziness q (which merely rescales time, $\tau_c \rightarrow \tau_c/q$, and drops out of the q -reduced threshold). It is destroyed only by placing the source on a symmetry node (dark modes $G_j = 0$), by source coalescence (rank drop), by moving the source adjacent to the absorbing boundary, or by pushing the start onto the shortcut, in which case the early peak escapes to $\tau = 0$ and leaves the interior fold theorem.

E. Direct simulation and channel resolution

Three checks confirm that the fold is a physical mechanism rather than an artifact of the spectral representation [Fig. 3]. (i) *Direct simulation*. A direct Monte-Carlo of the lazy ring, using no spectral input, reproduces the exact FPT density across four decades in τ [Fig. 3(a)], with the simulated splitting probability $\pi_{sc} = 0.565$ matching the exact value to 10^{-3} . (ii) *Channel resolution*. Because the absorption vector splits exactly into a shortcut entry and diffusive entries, the density decomposes as $f = f_{sc} + f_{diff}$ [Fig. 3(b)]; the early mode is shortcut capture and the late mode is diffusive return, so the

saddle-node is precisely the annihilation of the diffusive-return peak as the capture channel strengthens. This identifies which physical channel produces which arrival-time mode. (iii) *Non-antipodal robustness*. The same fold occurs off the reflection axis—e.g. at $\theta = 0.38$ across $b_c(\theta) \simeq 2.16$ [Fig. 3(c)]—so the mechanism is not special to the antipodal ring.

V. PHYSICAL REALIZATION AND NON-HERMITIAN CHARACTER

A. Embedding

Physically the shortcut is a localized target-delivery channel. From $p'(t) = Q_0 p - b|u\rangle\langle u|p$ the absorption current is $f(t) = f_0(t) + b p_u(t)$: while the searcher is at u it is irreversibly delivered to the target at rate b . In the survival sector this is a local killing term $-b|u\rangle\langle u|$, not a transport edge, and a finite gate of width ℓ with local trigger rate κ coarse-grains to a δ -sink of strength $\sim \kappa\ell$; the dimensionless b is $\kappa\ell$ normalized by the diffusive hop scale. The sharp physical statement is that a single localized delivery channel creates an *additional arrival-time scale*—a second peak in the full FPT density—with a predictable threshold $b_c(\theta)$.

B. Realizations

We rank candidate realizations by cleanliness. (i) *Feedback-controlled colloidal or active-particle search* is the strongest: a narrow gate at $x = \theta L$ (optical tweezers, a microfluidic pulse, or active-particle feedback) that, upon entry, occasionally delivers the particle to the target and ends the trial; b is the per-frame trigger rate. Releasing the particle at the gate ($\xi = \theta$, the convention under which $b_c(\theta)$ is quoted), recording first-arrival times over many trials, and scanning b at fixed θ —best near $\theta \simeq 0.38$, where b_c is smallest and the effect largest—should reveal the second peak for $b < b_c(\theta)$ and its annihilation at b_c , with the fold collapse (13) as the sharp signature. We caution that standard resetting-to-origin is *not* this problem: it reinjects probability into the survival domain, whereas the correct protocol is delivery-and-terminate. (ii) *Engineered or small-world transport networks* with a one-way express edge to an absorbing node realize $-b|u\rangle\langle u|$ directly (microfluidic maze with a one-way valve; programmable packet-routing graph; electrical random-walk analog with a tunable shunt to ground; lossy photonic link); two such edges provide the cleanest test of the rank-two triple peak and cusp. (iii) *Intracellular motor capture* to a reaction site realizes it in a coarse-grained, ideally reconstituted, in-vitro setting. In each case the caveat is that only a genuinely one-way, terminating channel maps to the rank-one killing defect.

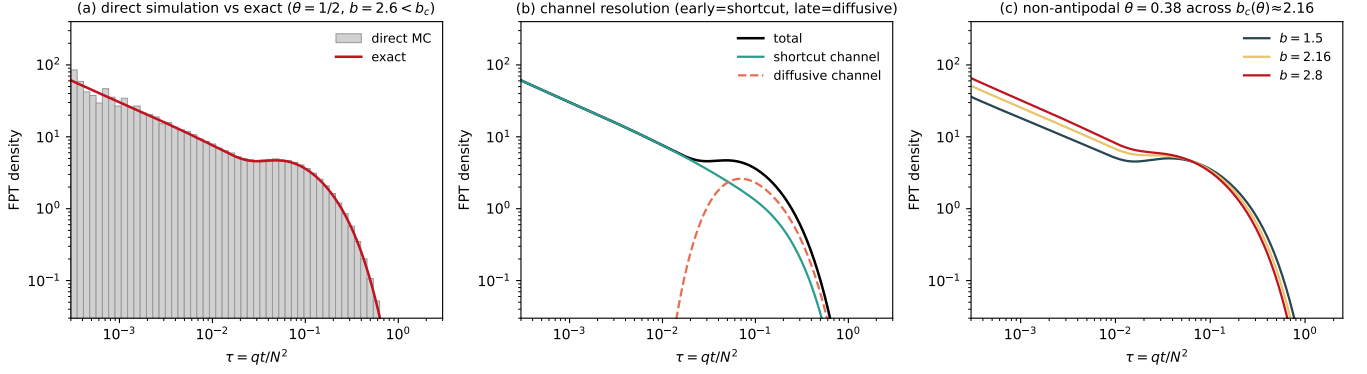


FIG. 3. The fold is a physical mechanism, not an artifact of the exact solution. (a) A direct Monte-Carlo simulation of the lazy ring (grey; 4×10^5 walkers, no spectral input) reproduces the exact FPT density (red) across four decades in τ ; the simulated splitting probability $\pi_{sc} = 0.565$ matches the exact value to 10^{-3} . (b) The absorption vector splits exactly into a shortcut entry (λ at u) and diffusive entries ($q/2$ at the neighbors of v), so the density decomposes as $f = f_{sc} + f_{diff}$: the early arrival-time mode is shortcut capture (monotone, dominant at small τ) and the late mode is diffusive return (a bump at $\tau \sim 0.05$). (c) The same valley-peak fold occurs away from the antipode, at $\theta = 0.38$ across $b_c(\theta) \simeq 2.16$. ($N = 100$, start $\xi = \theta$.)

C. Non-Hermitian character

We stress what the non-Hermitian character is and is not. Q_b is a passive open-Markov loss defect, analogous to an effective non-Hermitian Hamiltonian with a localized absorber (imaginary potential) [12]; it is *not* PT-symmetric, since there is no balanced gain and loss. We stress that the survival-sector operator is itself self-adjoint in the one-defect model (the directedness enters only the absorption vector), so “non-Hermitian” here denotes open-system probability loss, not a spectral mechanism. Crucially, b_c is *not* an exceptional point: an EP requires eigenvalue coalescence with a defective (Jordan) block, whereas here the eigenvalues remain real, simple and smooth through b_c (verified across $b = 2.9 \rightarrow 3.3$ at $N = 200$, consecutive gaps bounded away from zero), and the fold arises from cancellation among at least three signed residues, not from spectral degeneracy. The $(b_c - b)^{1/2}$ scaling in (13) is a saddle-node of the scalar $f(t; b)$, not EP eigenvalue splitting—the same exponent for a different object. Indeed, in the reversible diffusive limit the δ -sink operator is self-adjoint (Sturm–Liouville), so no spectral EP exists at all.

VI. DISCUSSION

We have shown that shortcut-induced FPT bimodality, long known phenomenologically, is the observable signature of an exactly solvable saddle-node bifurcation: a rank-one directed defect produces a second first-passage peak that exists precisely for $0 < b < b_c(\theta)$, is born and annihilated through a fold with universal $(1/2, 3/2)$ scaling, requires at least three sign-alternating modes, generalizes to a cusp for two defects, and survives in two dimensions. The single most defensible novelty is the bifurcation-theoretic classification with a predictive,

computable threshold, distinct from the known bimodality and from the underlying defect-resolvent method. The predicted fold exponents and the two-shortcut triple peak are directly measurable, and the cleanest platforms are feedback-controlled colloidal search and engineered transport networks.

Beyond the results proved and verified here, the analysis leaves a small set of technical questions. A fully rigorous $N \rightarrow \infty$ statement requires, in addition to the standard convergence of fixed-mode residues (App. B), uniform C^1 convergence of the moment sums $S_{1,2,3}$ in a neighborhood of the fold; we state the precise hypotheses and support them by the observed $O(N^{-2})$ fold convergence, but do not prove uniform C^1 control of the tail. The large- L renormalized scaling of b_c in two dimensions, and whether the constants $0.7890\dots$ and the antipodal b_c admit elementary closed forms, remain open. None of these affects the exact finite-system results or the numerically certified continuum thresholds.

ACKNOWLEDGMENTS

We thank colleagues at the Bristol Centre for Complexity Sciences for discussions.

Appendix A: Exact finite- N determinant and residues

We derive Eqs. (2)–(3). Let P_0 be the $(N-1) \times (N-1)$ transient (sub-stochastic) transition matrix of the lazy ring with $v = 0$ deleted; on the interior sites it is a symmetric tridiagonal path with diagonal $1 - q$ and off-diagonal $q/2$. The directed shortcut diverts $\lambda = \beta(1 - q)$ from the self-loop at u into the absorbing target, i.e. it subtracts λ from the (u, u) entry: $P_b = P_0 - \lambda |u\rangle\langle u|$, the

rank-one defect of (1). The first-passage generating function to v from a start r , $\bar{F}_r(z) = \sum_{t \geq 1} F(t)z^t$, is a ratio of resolvent elements of P_b ; by the matrix-determinant lemma its poles are the zeros of

$$\det(I - zP_b) = \det(I - zP_0) [1 + z\lambda G_{uu}^0(z)], \quad (\text{A1})$$

with $G^0(z) = (I - zP_0)^{-1}$. Introducing the spectral variable $y = 1 + (z^{-1} - 1)/q$ maps the lazy path onto the standard second-difference operator: $\det(yI - \frac{1}{2}A_{\text{path}}) = 2^{-(N-1)}U_{N-1}(y)$ and, by the cofactor structure of a tridiagonal inverse,

$$G_{uu}^0 \propto [(yI - \frac{1}{2}A_{\text{path}})^{-1}]_{uu} = \frac{2U_{u-1}(y)U_{N-u-1}(y)}{U_{N-1}(y)}. \quad (\text{A2})$$

Substituting (A2) into (A1) and clearing U_{N-1} gives the characteristic determinant $D_u(y) = aU_{N-1}(y) + 2U_{u-1}(y)U_{N-u-1}(y)$ of (2), with $a = q/\lambda$. The residue of $\bar{F}_r$ at each root y_j produces the geometric sum (3) with $s_j = 1 - q + qy_j$ and $B_j = q\mathcal{N}_{r,u}(y_j)/D'_u(y_j)$. Writing $A_m := U_{m-1}(y)$ (with the convention $A_0 = U_{-1} = 0$), the piecewise numerator is

$$\mathcal{N}_{r,u} = a[A_r + A_{N-r}] + 2A_{N-u}[A_r + A_{u-r}], \quad r \leq u, \quad (\text{A3})$$

and its mirror ($r \leftrightarrow N - r$, $u \leftrightarrow N - u$) for $r \geq u$. Finally, by Sherman–Morrison the eventual channel (splitting) probability—the fraction of trajectories absorbed through the shortcut—is

$$\pi_{\text{sc}}(r) = \frac{2 \min(r, u) [N - \max(r, u)]}{aN + 2u(N - u)}, \quad (\text{A4})$$

which for the antipodal case reduces to $\pi_{\text{sc}} = \rho/(a + L)$ with $\rho = \min(r, N - r)$ the start's short-arc distance to the absorbing target. We emphasize that π_{sc} is the eventual splitting probability, *not* the mass under the first peak; conflating the two is a common error we deliberately avoid.

Appendix B: Continuum amplitude from the defect resolvent

We derive Eqs. (7)–(8) from the resolvent of $H_b = -\frac{1}{2}\partial_x^2 + b\delta_\theta$ on $(0, 1)$ with Dirichlet ends, continued to $z = -k^2/2$ (so $\mu = k^2/2 = 2w^2$).

1. *Free resolvent.* With $\varkappa = \sqrt{2z}$ (a spectral variable, not the laziness q), $R_0(x, y; z) = 2 \sinh(\varkappa x_{<}) \sinh(\varkappa(1 - x_{>}))/(\varkappa \sinh \varkappa)$; continued to $\varkappa = ik$,

$$r_0(x, y; k) = \frac{2 \sin(kx_{<}) \sin(k(1 - x_{>}))}{k \sin k}. \quad (\text{B1})$$

2. *Defect resolvent (Sherman–Morrison, rank-one δ -sink).*

$$r_b = r_0 - \frac{b r_0(x, \theta) r_0(\theta, y)}{1 + b r_0(\theta, \theta)} = r_0 - \frac{4b \phi(x) \phi(y)}{k \sin k D_\theta}, \quad (\text{B2})$$

with $\phi = \phi_{k, \theta}$ of (8); the pole condition $1 + b r_0(\theta, \theta) = 0$ is exactly $D_\theta(k; b) = 0$ of (5). 3. *Eigenfunction.* The residue eigenfunction is $\psi_j = \phi_{k_j, \theta}$: it is continuous at θ , satisfies the Dirichlet conditions, and obeys the δ -sink jump $\psi'(\theta^+) - \psi'(\theta^-) = 2b\psi(\theta)$, which is algebraically equivalent to $D_\theta = 0$ (verified to 1.8×10^{-14}). 4. *Residue = FPT amplitude.* The survival transform is $\bar{Q}_\xi(z) = \int_0^1 r_b(y, \xi; z) dy$ and $\bar{f} = 1 - z\bar{Q}$; the residue at $z = -\mu_j$ (using $k - k_j = -(z + \mu_j)/k_j$) gives the normalization-independent amplitude

$$G_j = \mu_j \psi_j(\xi) \frac{\int_0^1 \psi_j}{\int_0^1 \psi_j^2} = \mu_j \frac{\phi_{k_j, \theta}(\xi) I}{J}, \quad (\text{B3})$$

i.e. Eq. (7) with $\mu_j = 2w_j^2$. 5. *Closed forms.* With $A = \sin k\theta$, $B = \sin k(1 - \theta)$,

$$I(k, \theta) = \frac{1}{k} [B(1 - \cos k\theta) + A(1 - \cos k(1 - \theta))], \quad (\text{B4})$$

$$J(k, \theta) = \frac{1}{2} [\theta B^2 + (1 - \theta) A^2] - \frac{AB \sin k}{2k}, \quad (\text{B5})$$

matching direct quadrature to $\sim 10^{-9}$. 6. *Normalization identity.* At a root of (5),

$$J(k_j, \theta) = \frac{\sin(k_j) D_k(k_j; b)}{4b}, \quad (\text{B6})$$

with $D_k = \sin k + k \cos k + 2b[\theta \cos(k\theta) \sin(k(1 - \theta)) + (1 - \theta) \sin(k\theta) \cos(k(1 - \theta))]$; this exact algebraic identity (verified to 2.9×10^{-15}) ties the residue norm to the determinant derivative, and $D_k(k_j) \neq 0$ certifies each affected mode as nondegenerate. 7. *Antipodal reduction.* For $\theta = \frac{1}{2}$, $G_{\xi, 1/2} = 4w(1 - \cos w) \sin(2w\xi^*)/[\sin^2 w(1 + b(b + 2)/4w^2)]$ with $\xi^* = \min(\xi, 1 - \xi)$, recovering the previously verified antipodal amplitude. 8. *Node modes.* When $\sin(n\pi\theta) = 0$ the mode decouples from the sink and is the free Dirichlet mode $\sin(n\pi x)$, with amplitude $G_n^{\text{node}} = n\pi[1 - (-1)^n] \sin(n\pi\xi)$ (even n vanish, and all node modes vanish at $\xi = \theta$). The reconstructed master curve (affected plus node) matches exact finite- N residues to $\sim 10^{-5}$ with $O(N^{-2})$ convergence.

The $N \rightarrow \infty$ limit. Convergence of fixed-mode residues $G_{N,j} \rightarrow G_j$ follows from norm-resolvent convergence of the killed generator under standard hypotheses (A) $\xi_N \rightarrow \xi$, $\theta_N \rightarrow \theta$; (B) bulk convergence $H_N \rightarrow -\frac{1}{2}\partial_x^2$; (C) δ -sink form scaling $V_{N,\theta} \sim bN$; (D) a symmetrizable transient generator; (E) simple isolated poles. Passing the fold (10) to the limit additionally requires (F) uniform C^1 convergence of $S_{1,2,3}$ (including ∂_b) on a neighborhood $\tau \in [\tau_c/2, 3\tau_c/2]$ of the fold—not down to $\tau = 0$, where the heat-kernel short-time singularity lives—for which a sufficient practical bound is $\mu_{N,j} \geq cj^2$, $|G_{N,j}| \leq Cj^r$ uniformly. Under (F), fold persistence is a consequence of the implicit-function theorem: the Jacobian $\det DF_c = +S_{1,b}S_3$ is nonzero by nondegeneracy, giving a unique nearby finite- N fold with $b_{c,N} - b_c = O(\varepsilon_N)$, consistent with the observed $O(N^{-2})$. We state (F) as a hypothesis, supported by the numerics, rather than a theorem.

Appendix C: Fold criterion: proofs and asymptotics

Fold criterion. From $\Phi = \sum_j G_j e^{-\mu_j \tau}$, $\Phi' = -\sum_j G_j \mu_j e^{-\mu_j \tau} = -S_1$ and $\Phi'' = \sum_j G_j \mu_j^2 e^{-\mu_j \tau} = S_2$, so an interior extremum is $S_1 = 0$ and two extrema merge at $S_1 = S_2 = 0$, Eq. (10). The parameter derivative of a root of (5) is $\partial_b k_j = -D_b/D_k$ with $D_b = 2 \sin(k\theta) \sin(k(1-\theta))$ and D_k as in (B6), which propagates into $\partial_b S_1$ and gives the nondegeneracy condition needed for a structurally stable fold.

Minimal-mode theorem. As in Sec. IIID, $S_1 = S_2 = 0$ has no two-mode solution with $A_j \neq 0$; the minimal count is three, with the alternating-sign ratio $A_1 : A_2 : A_3 = (\mu_3 - \mu_2) : (\mu_1 - \mu_3) : (\mu_2 - \mu_1)$. This retires an earlier (incorrect) two-mode heuristic and shows the fold is intrinsically a many-mode cancellation.

Endpoint law. As $d = \min(\theta, 1-\theta) \rightarrow 0$ the fold is governed by a half-line boundary-layer problem (absorbing end at 0, δ -sink at rescaled position $y = 1$ of strength B , start $y = 1$); its saddle-node occurs at the universal constant $B^* = 0.7890261736$, giving $b_c(\theta) \sim B^*/d$ and $\tau_c \sim 0.1579221011 d^2$, Eq. (11). Near the antipode a regular perturbation of $\tan w = -2w/b$ yields $b_c(\frac{1}{2} + \epsilon) = 3.0764323604 - 133.107 \epsilon^2 + O(\epsilon^4)$, with the antipodal values (12) obtained from $S_1 = S_2 = 0$ directly ($S_3 = 2.83 \times 10^5$, $\partial_b S_1 = 21.66$, nondegenerate). The truncated thresholds $b_c^{(3)} = 3.046$, $b_c^{(5)} = 3.0764$ confirm rapid mode convergence.

Appendix D: Two-shortcut determinant and cusp

For two sources the secular condition is $\det[I_2 + BW(k)] = 0$ with $W_{ab} = r_0(k; \theta_a, \theta_b)$ the interval resolvent (B1); expanding the 2×2 determinant gives $D_{12}(k)$ of (15), whose cross term $\propto b_1 b_2$ is the genuine two-defect interaction (it $\rightarrow 0$ as $\alpha \rightarrow \eta$, recovering a single defect of strength $b_1 + b_2$). The rank-two residue eigenfunction is $\psi_j(x) = -\sum_a b_a c_a^{(j)} r_0(k_j; x, \theta_a)$ with $c^{(j)} \in \ker[I_2 + W(k_j)B]$, and $G_j = \mu_j \psi_j(\xi) \int \psi_j / \int \psi_j^2$ with $\int \psi_j = -\sum_a b_a c_a I_{\theta_a}$ and $\int \psi_j^2 = \sum_{ab} b_a c_a b_b c_b C_{ab}$; the constituents I_θ [Eq. (B4)] and C_{ab} are elementary. The cusp solves $S_1 = S_2 = S_3 = 0$ with $S_4 \neq 0$ and

a nonzero unfolding Jacobian $\det \partial(S_1, S_2)/\partial(b_1, b_2)$. At the located point (16) an independent reimplementing gives normalized residuals $\tau S_1/\Phi \approx 7 \times 10^{-12}$, $\tau^2 S_2/\Phi \approx 5 \times 10^{-11}$, $\tau^3 S_3/\Phi \approx 4 \times 10^{-10}$, $\tau^4 S_4/\Phi \approx -1.67$, and $\det \partial(S_1, S_2)/\partial(b_1, b_2) \approx -6.9 \times 10^7$, confirming a nondegenerate codimension-two cusp.

Appendix E: Numerical methods and reproducibility

Exact finite- N FPT poles are computed from the symmetric transient block via `numpy eigvalsh`, with amplitudes from the closed-form residues; high-precision cross-checks use `mpmath`. Continuum spectra are the roots of (5)/(15), and amplitudes use (7) and App. D. All scripts are deterministic. Table I collects the verification tolerances quoted throughout. The figures are generated by `dpma_prr_figures.py` (results) and `dpma_schematic.py` (model); the existence boundary by `dpma_saddle_node_bc_theta.py`; the amplitude/master curve by `dpma_general_u_master_curve.py`; the fold scaling by `dpma_saddle_node_certification.py`; the cusp by `dpma_cusp_verify.py`; the rank-two determinant and triple peak by `dpma_multishortcut.py`; the 2D test by `dpma_2d_universality.py`; and the direct-simulation, channel-resolution, and non-antipodal panels by `dpma_channel_mc.py`. Over the scanned range we find the double-peak set to be a single connected interval in b (no re-entrant window); the 2D fold is computed at $L = 31$.

TABLE I. Independent numerical verifications (this work).

Quantity	Agreement
δ -sink jump $= D_\theta$ [Eq. (5)]	1.8×10^{-14}
I, J vs. quadrature [Eqs. (B4),(B5)]	$\sim 10^{-9}$
Identity $J = \sin(k) D_k / 4b$ [Eq. (B6)]	2.9×10^{-15}
Master curve vs. exact residues ($N = 1200$)	$\sim 10^{-5}$
$b_c(\theta)$ vs. independent numerics	$\sim 10^{-6}$
Endpoint constant $b_c \cdot d \rightarrow B^*$	0.7890
Rank-2 secular $D_{12}(k_j)$ ($N = 800$)	$\sim 10^{-5}$
Cusp residuals $\tau^n S_n / \Phi$ ($n \leq 3$)	$\lesssim 10^{-9}$

-
- [1] S. Redner, *A Guide to First-Passage Processes* (Cambridge University Press, Cambridge, 2001).
 - [2] R. Metzler, G. Oshanin, and S. Redner (eds.), *First-Passage Phenomena and Their Applications* (World Scientific, Singapore, 2014).
 - [3] O. Bénichou, C. Loverdo, M. Moreau, and R. Voituriez, *Rev. Mod. Phys.* **83**, 81 (2011).
 - [4] E. W. Montroll and G. H. Weiss, *J. Math. Phys.* **6**, 167 (1965).
 - [5] L. Giuggioli, *Phys. Rev. X* **10**, 021045 (2020).
 - [6] A. Godec and R. Metzler, *Universal Proximity Effect in Target Search Kinetics in the Few-Encounter Limit*, *Phys. Rev. X* **6**, 041037 (2016).
 - [7] T. G. Mattos, C. Mejía-Monasterio, R. Metzler, and G. Oshanin, *Phys. Rev. E* **86**, 031143 (2012).
 - [8] E. Almaas, R. V. Kulkarni, and D. Stroud, *Scaling properties of random walks on small-world networks*, *Phys. Rev. E* **68**, 056105 (2003).
 - [9] D. S. Grebenkov, *Paradigm Shift in Diffusion-Mediated Surface Phenomena*, *Phys. Rev. Lett.* **125**, 078102 (2020).

- (2020).
- [10] P. C. Bressloff, *Diffusion-mediated absorption by partially-reactive targets: Brownian functionals and generalized propagators*, J. Phys. A **55**, 205001 (2022).
- [11] V. I. Arnol'd, *Catastrophe Theory*, 3rd ed. (Springer, Berlin, 1992).
- [12] Y. Ashida, Z. Gong, and M. Ueda, Adv. Phys. **69**, 249 (2020).